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# Artificial Intelligence Recommendations Amplify the Sharing of True and Fake News on Social Media by Appealing to Fast Cognition

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## ABSTRACT

Many news stories that users see on social media are recommended by artificial intelligence (AI), but most users are unaware of this. Policymakers have argued that increasing transparency about the role of AI may reduce users' sharing of fake news. We use social influence theory to argue that such explicit labeling of news stories may have the opposite effect of what policymakers intend by triggering users to rely on fast System 1 cognition rather than more effortful rational System 2 cognition. We conducted three experiments with two different forms of AI in United States. The results of Study 1 show that labeling stories as recommended by an AI agent had similar effects to labeling them as recommended by a human expert: it encouraged users to use fast cognition, which made them more likely to share fake news (as well as true news). Study 2 examined the effects of using a more machine-like algorithmic depiction of AI and found that, once again, AI labels made users more likely to share fake news, which was validated again in Study 3 in the post-Gen AI era. Our research contributes to theory by showing that labeling social media stories as recommended by AI serves as a signal of positive social influence that triggers the use of fast cognition instead of rational cognition and thereby exacerbates rather than mitigates the spread of fake news.

## KEYWORDS

social media; online recommenders; AI recommendations; fake news; fast cognition; social influence; dual process cognition; signaling; system-1 cognition

## Introduction

Fake news has become a major societal problem, with its effects being felt in numerous elections around the world [3, 31, 34, 64, 77]. The U.K. House of Commons report called fake news a “crisis of democracy” due to its use in disinformation campaigns designed to spread hate and discord [37]. Fake news stories can distort users' understanding [31] because social media users are typically not motivated to expend cognitive effort to determine whether a story is true or false [31, 64, 77]. Instead, they often rely on experiential System 1 cognition (which is fast and automatic) rather than rational System 2 cognition

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(which is deliberate and effortful), a pattern that plays a central role in the spread of fake news [8, 64, 77].

Social media platforms use artificial intelligence (AI) to recommend news stories to users [7], but most users do not realize this [48, 57, 103]. Experts have recommended that users should be informed about the role of AI [83, 105], arguing that such knowledge would reduce the spread of fake news by triggering users to devote more attention to a news story before sharing it [20, 88]. As a result, policymakers in many countries are considering laws requiring social media platforms to label content recommended by AI. For example, China has implemented a law and the European Parliament is considering new laws requiring social media platforms to disclose the use of AI in content recommendations (see Online Supplemental Appendix 4). These regulatory initiatives are slightly different, but all have, at their core, a belief that labeling news stories as recommended by AI will encourage users to think more carefully about them by triggering users' feelings of algorithmic aversion [82, 99], thereby reducing users' sharing of fake news [4, 62]. However, a recent study shows that AI recommendations *increase* the belief and sharing of fake news [55], suggesting algorithm appreciation, not algorithm aversion, influences social media use. This initial study [56] was conducted before the rise of generative AI (GenAI), and there is evidence of aversion to AI-created products [82]. Two critical theoretical questions that also have practical relevance are why do AI labels have these effects and do the findings of the initial study still hold in the GenAI era.

In this paper, we use social influence theory [17] to argue that labeling stories as recommended by AI has a similar informational social influence effect as a recommendation by a human expert and triggers users to use fast System 1 cognition rather than more effortful System 2 cognition, which *increases* rather than decreases the sharing of fake news. A reliance on fast and intuitive System 1 cognition, rather than the use of slower and more rational System 2 is what Pennycook and Rand [77] call "lazy cognition." Therefore, instead of prompting greater use of System 2 rational cognition as policymakers hope, we argue that AI labels may actually prompt users to rely on fast cognition, which will lead to greater sharing of fake news [64, 78]. Our research question is:

*Research Question (RQ): Does disclosing that social media news stories are recommended by AI influence social media users' cognition and their likelihood of sharing fake news stories?*

We conducted three online experiments investigating the extent to which AI labels affected users' likelihood of sharing news stories on social media. We studied the two forms of AI commonly deployed on the Internet: visible agents and embedded algorithms [32]. Study 1 (conducted before the rise of GenAI) examined the effects of AI recommendations by comparing them to no recommendations and to human experts' recommendations. The findings show that labeling news stories with AI presented as a humanlike agent had similar effects as labeling stories as recommended by a human expert: it encouraged users to rely more on fast System 1 cognition than rational System 2 cognition and as a result, they were more likely to share fake news. Study 2 (conducted during the rise of GenAI) found the same positive effects on fast cognition and fake news sharing using an embedded algorithm depiction of AI. Study 3 (conducted post GenAI) validated the findings of Study 2.

Our research makes four important contributions (two theoretical and two practical). First, it shows that one underlying theoretical mechanism for algorithm appreciation is user cognition. When a story was labeled as recommended by AI, users

tended to rely on fast cognition, increasing their reliance on experiential System 1 cognition rather than rational System 2 cognition. This suggests that algorithm aversion does not generalize to social media news recommendations. Second, the two different AI designs (a more humanlike AI agent versus an AI algorithm) had similar effects on user cognition and news-sharing behavior, although the more humanlike AI had stronger effects. This shows that AI has social influence even when it is an embedded algorithm, not just as a human-like agent, which extends Computers Are Social Actors (CASA) framework [66, 67] beyond its prior bounds as a theory for conversational agents and also provides supporting evidence for AI impacts on human mind [12]. From a practical perspective, our results provide clear guidance to policymakers: requiring platforms to label the role of AI in recommending social media news stories will likely *increase* the spread of fake news. Therefore, policymakers should be very careful about policies to increase transparency about AI's role on social media. We also found that AI labels had little effect or actually increased the sharing of true news, so the implication for platforms seeking to increase user engagement is to embrace the role of AI and promote their use of AI in curating content, because this is likely to increase engagement.

## Prior Theory and Research

### *Fake News on Social Media*

The proliferation of news on social media has changed the news environment. About one quarter of American adults often get their news from social media with another one-third sometimes using social media for news [79]. Social media differs from traditional news in three aspects. First, social media presents a mixture of content generated by users and professionals [69]. Unlike traditional media channels, much of the content on social media is published without any quality control [91]. Second, social media users usually do not actively seek news content by selecting stories themselves; instead, news exposure is more likely to be incidental, moderated by algorithms that select stories from companies and individuals the user has previously engaged with, as well as paid advertisements [2]. Third, social media provides features for easy news engagement (e.g., share, comment, and like) [14]. These three aspects provide an interactive social news environment and have also enabled fast and wide diffusion of fake news [10, 51, 53], which can harm individuals (e.g., health) [98] and society as a whole (e.g., partisanship) [71, 95].

The seminal definition of fake news is “news articles that are intentionally and verifiably false and could mislead readers” [3:213]. This definition has been widely adopted in prior research [36, 45, 46, 62]. We extend the research on fake news to understand the reasons why people fail to recognize fake news. Past research has focused on news content (e.g., emotionality) [36, 90], individual motivations (e.g., altruism, self-presentation) [21, 31, 52], and reducing cognitive effort (e.g., confirmation bias) [64]. People are more likely to believe news stories that align with their preexisting opinions [62, 73]; therefore, they are more likely to share news stories that align with their preexisting beliefs and opinions [64]. For the fake news actions, early fake news detection can be conducted, and suspicious content will be removed [104]. Facebook once put warnings on fake news [62], but this has changed.

Social Media and AI

Social media has long used AI and algorithms to curate content and select what posts users see [101]. The problem of fake news was initially ascribed to AI helping to spread fake stories, until research showed that humans are at least as much to blame [101]. AI and algorithms do exacerbate the problem of fake news by creating echo chambers by recognizing what content users prefer and then showing them more content like that [28]. This has led to different users being shown very different stories about the same events [28].

Despite the prevalence of AI, users are generally unaware that AI recommends the social media stories they see [48, 103]. As recently as 2023, Pew Research Center found that “44% of Americans think they do not regularly interact with AI” [44]. We verified these prior findings in April 2025. We recruited 100 participants from Prolific Academic and asked whether AI was used to select and filter advertisements and posts from friends using a 1-5 scale with items adapted from recent studies (1 = strongly disagree, 3 = unsure, 5 = strongly agree; see Online Supplemental Appendix 3). We coded a mean response of less than 3.00 as no AI involvement, 3.00-3.99 as unsure, and 4.00 or more as AI involvement. We found that most people were aware of AI recommendations for advertisements, with 63% of the participants stating they believed AI was involved in selecting the advertisements they saw, versus 30% who were unsure, and 7% who believed AI was not involved. However, only 33% of the participants thought friends’ posts were curated by AI versus 38% unsure and 29% believing AI was not involved (Table 1). Our findings align with prior studies in showing that only a minority of users are aware of the role of AI in determining which posts from other users they see [48, 103].

This lack of awareness has prompted some experts and policymakers to argue that the role of AI should be explicitly disclosed [83, 105]. The key question is: how will users react to such disclosures? This becomes especially important as social media firms that use AI are likely to present the use of AI in the best possible light. We have a preliminary understanding of how AI recommendations influence human behaviors in areas such as health-care [40], marketing [55], or consumer behaviors [33]; however, there is no research to guide us on the impacts of AI on individuals’ cognition and news sharing behavior in social media.

Cognition and Fake News

One theoretical explanation for users’ belief in and sharing of fake news is a lack of reasoning—social media users do not devote cognitive effort to thinking about the truthfulness of the news stories they share [10, 62, 74, 77]. In other words, the reason why confirmation bias plays such a central role in fake news is that users do not exert the cognitive effort needed to overcome the initial belief in fake news that aligns with their

Table 1. Awareness of AI involvement.

|                | Believe there is<br>no AI involvement (%) | Unsure<br>(%) | Believe there is<br>AI involvement<br>(%) |
|----------------|-------------------------------------------|---------------|-------------------------------------------|
| Advertisements | 7                                         | 30            | 63                                        |
| Friends’ Posts | 29                                        | 38            | 33                                        |

Abbreviation: AI, artificial intelligence.

preexisting opinions [63]. As a result, users tend to believe and share fake news stories that support their existing beliefs [62, 77]. Many dual cognition models have been proposed, with different authors using different terms, but with substantial theoretical agreement about the two types of cognition [26, 27, 93]. One type of cognition is experiential [23], impulsive [96], intuitive [27, 35], associative [92], and fast System 1 [41], while the other is rational [23], reflective [96], rule-based [92], deliberate/analytic [27, 35], and slow System 2 [41].

We use the terminology of Epstein [24] and Kahneman [41], and describe them as experiential System 1 cognition and rational System 2 cognition. *Experiential System 1 cognition*, is emotional, effortless, quick, unconscious, rapid to execute, and the primary style used for normal daily activities (e.g., recognizing an acquaintance, stopping when the traffic light is red) [24, 27, 41]. In contrast, *rational System 2 cognition*, is slow, analytical, conscious, and initiated actively with people's awareness and control [24, 27, 41]. Experiential System 1 cognition executes automatically, while rational System 2 cognition is typically only invoked when there is a reason to devote effort to making a decision [24, 25, 27, 41]. As a result, the focus is on the extent to which individuals rely on System 1 cognition versus choosing to engage in and relying on System 2 cognition when making decisions [24, 25, 41].

System 1 and System 2 cognition have been a consistent interest to scholars of social media [27]. The extent to which users rely on System 1 versus System 2 cognition has a strong influence on the belief and sharing of fake news [8, 64, 77]. When someone relies more on experiential cognition, they are making decisions by instinct, resulting in quick and impulsive choices [60, 68, 77]. When someone relies more on rational cognition, they realize that the situation deserves deeper consideration, resulting in deeper thought [60, 68].

A reliance on System 1 cognition, called "lazy cognition" by Pennycook and Rand [77], is a central reason why social media users believe and share fake news [62, 64, 73]. When users engage in more effortful System 2 cognition, belief in fake news stories is reduced, and as a result, they are less likely to share them [62, 63, 77]. There is a strong relationship between the reliance on System 1 cognition and confirmation bias because overcoming the initial confirmation of one's pre-existing beliefs requires cognitive effort; therefore, people are more likely to believe fake news stories aligned with their opinions due to the ease of using the experiential cognition, and thus fail to use rational System 2 cognition [62, 77]. Thus, understanding why people have an over-reliance on System 1 and fail to rely on rational System 2 cognition is an important step in reducing the spread of fake news [62, 77].

While System 1 and System 2 cognition can be negatively correlated [68], they are not *always* negatively correlated [94]. Sojka and Giese [94] categorized their participants into four groups depending upon the extent to which they reported relying on one or both to make their decisions. Our theorization centers on the use of fast cognition (which Pennycook and Rand [77] call "lazy cognition") where people rely more on System 1 cognition than System 2 cognition [60, 68]. Therefore, we define fast cognition as the extent to which users rely more on System 1 cognition than on System 2 cognition [60, 68].

### **Social Influence of Recommendations**

On social media, news stories are selected by the platform's AI, not the users, so users have limited control over the source of the stories they see [64]. The source that created the story

may not be recognized [45] therefore, users may instead focus on the person who recommended the story (i.e., the recommender) [69, 91] as the recommender is usually displayed more prominently than the source [45]. When encountering a story recommended by another person, a user may share it, which signals appropriateness and positive assessment [50]. Past research suggests that social media users are susceptible to the influence of others [59], and, as a result, the source or recommender of a story can affect its spread [31, 45]. On social media, experts and friends can be deliberately “followed” to obtain important information, whereas AI is used ubiquitously, regardless of whether users are aware of it or not.

One of the most important drivers of human behavior is the social influence of those around them [17, 18, 102]. Social influence can be either informational (serving a desire for accurate decisions) or normative (serving a desire to build or maintain interpersonal relationships) [17, 49]. Experts have informational influence (people follow the views of experts because of their professional knowledge [80]).

### **Expert Recommendation**

News organizations have leveraged the power of social media platforms to promote their content, reach a broader audience, and generate profits [69]. News stories recommended by these experts appear on the newsfeeds of users when they follow or “friend” an expert. Experts exert a strong influence on people’s attitudes, beliefs, decision-making, and actions. People tend to follow experts as authorities to inform their own choices [70, 87]. Political opinions can influence the credibility of experts [75], with liberals being more likely to rely on experts than conservatives [29].

### **AI Recommendation**

As in other domains, such as algorithm-based product and music recommendations, AI is commonly used to curate users’ news feeds on most social media platforms. Using historical data and user profiles as inputs, these algorithmic systems, as an embedded form of AI [32], can recommend content that fits users’ tastes. Recent studies show that people hold different levels of trust regarding AI, shaped by individual factors (e.g., demographics, familiarity), AI related factors (e.g., agency) [82, 100], and the context of human-AI interaction (e.g., culture, social influence), which, in turn, determines AI acceptance [54]. There is evidence of algorithm aversion, where people refuse to accept AI recommendations either because they believe AI to be less capable [40] or due to biases against AI [99]. This can be overcome by familiarizing oneself with AI [58, 99].

There is also evidence that users treat AI agents as social actors when they are deployed as conversational agents [30, 66, 67]. The Computers are Social Actors (CASA) framework [66, 67] posits that when interactions with AI conversational agents resemble human conversations, they subconsciously trigger the use of social scripts for human interaction, leading humans to anthropomorphize them [22, 61]. Users are aware that the AI agent is not a human, yet they subconsciously treat it as such, even though they often deny doing so. The subconscious nature of anthropomorphism means that users are usually not aware of their behavior [22, 30, 66, 67]. We follow this stream of research and theorize that recommendations by AI exert social influence.

## Research Model and Hypothesis Development

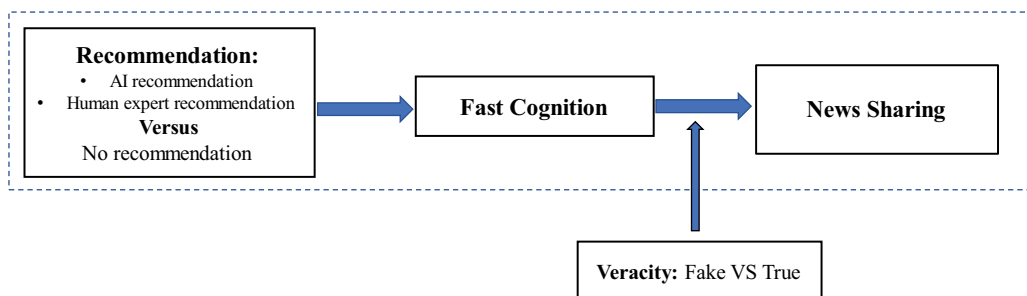
Our research model integrates social influence with dual process cognition to examine the impacts of two different types of news recommendations on individuals' cognition and fake news sharing. Many misbeliefs about fake news arise when experiential System 1 cognition produces an erroneous conclusion and the individual fails to engage in rational System 2 cognition to override it [65–67, 76, 81]. When people use fast cognition to assess social media news stories, they are more susceptible to confirmation bias and are more likely to believe and share news stories aligned with their preexisting opinions [65–67, 76, 81].

We theorize that the AI and human expert recommenders will have informational influence because users trust them to provide recommendations [60], and thus recommendations from AI and human experts will increase the use of fast cognition (i.e., people will rely more on experiential cognition than rational cognition). This, in turn, will increase news sharing of both true and fake news stories on social media (see Figure 1). Taken together, we propose that cognition will mediate the relationship between news recommendations and news sharing. We consider these in the following sections.

### Recommendation and Cognition

Experiential and rational cognition work together to guide behavior, with their relative strengths influenced by situational cues [17, 45]. We theorize that the extent to which a user relies on fast cognition (i.e., relying on experiential cognition to a greater extent than rational cognition) is influenced by who or what recommended a news story (i.e., a human expert or AI). When a user encounters a story, experiential System 1 cognition instantly and automatically produces a conclusion to share it or not [65, 76], influenced by the content of the story, the extent to which the content aligns with the individual's a priori beliefs (i.e., confirmation bias), and who recommended it [65, 76].

The conclusion produced by experiential cognition is assessed by a conflict monitoring process which determines the amount of rational System 2 cognition that will be invested [76]. Humans are cognitive misers who prefer to expend a minimal amount of effort [45, 101]; therefore, people often rely on System 1 for less important situations [17, 45]. Social media users often choose not to engage in effortful System 2 cognition when considering news stories, relying instead on System 1 cognition [65, 81], which increases the spread of fake news [67, 82].



**Figure 1.** The theoretical model.

Social influence can induce fast cognition when people rely on the source when making decisions, rather than investing effort in rational cognition [6]. For example, studies on persuasion have found that people often apply simple heuristics (e.g., accepting expert opinions), instead of considering the information in a deliberate, rational manner [84]. They adopt the views of experts because experts are perceived to be trustworthy and capable [60] and have access to resources that enable them to make accurate assessments [88]. Fast cognition is likely to dominate when the source of influence is perceived to be capable [1].

Policymakers believe that social media users exhibit algorithm aversion [102] and perceive AI to be a dubious source of news recommendations and thus will be less likely to use fast cognition when they encounter news stories recommended by AI. We argue the opposite: AI will be a source of informational social influence akin to a recommendation by a human expert. Therefore, in this study, we examine the impacts of the disclosure of AI recommendations on individuals' situational cognition and compare these effects with those of human expert recommendations.

### **Expert Recommendation**

Human experts are reliable sources of news, such as editors and reporters for newspaper and TV news, who have informational influence [106]. People tend to assume that expert recommendations are correct [1], because experts have professional knowledge, unique information, and high ability in a given area [88]. They can produce decisions and assessments with a low probability of errors, and they do a better job of judging the truthfulness of news stories than most social media users [49]. Therefore, a story recommended by an expert would be more likely to be perceived as true, having gone through a gatekeeping process by the experts using their professional knowledge and expertise [41]. Users will be less likely to feel the need to invoke rational cognition and more likely to rely on their experiential cognition. Thus, users will be more likely to perceive that they can rely on experts, and therefore rely on fast cognition. Therefore, we hypothesize:

*Hypothesis 1 (H1): Labeling a news recommendation as coming from a human expert will induce users to rely more on fast cognition, compared with no recommendation.*

### **AI Recommendation**

We theorize that AI will influence user cognition via two theoretical pathways. First, news is highly politically polarized in the United States, with many people believing that political bias influences social media users [49] and expert journalists alike [104]. People have been found to have the lay belief that AI is a neutral tool [42], has no abstract goals, uses only optimization to make decisions [20], and acts with little selfishness or personal intentions [33]. Research in other areas shows a high acceptance of AI recommendations for objective tasks [11], utilitarian products [58], and profile photos [106], but less so for recommending jokes [112], hedonic goods [58], and moral decision-making [20]. All this evidence indicates that users believe that AI is good at rational reasoning, and making minimal errors [42]. News recommendations should be an objective and fact-based task if not motivated by political bias or malicious intentions [13]. Hence, users should perceive that AI is reliable for news recommendation, and thus be less likely to invoke their own effortful rational cognition when presented with an AI recommendation.

Second, AI is widely adopted in various aspects of life, and most people are familiar with AI recommendations for music, videos, and other forms of media. [46]. AI recommendations usually align with individuals' preferences, foster positive experiences, and enhance satisfaction [24, 56]. Users are seldom provided with adequate information to cognitively understand and analyze AI's underlying processes [93]. Because it is difficult for users to understand how and why AI makes a recommendation, users tend to accept AI recommendations at face value [43]. This pattern of accepting AI recommendations using experiential cognition for other media applications (e.g., music, videos) may carry over into the social media context to produce a feeling of rightness when seeing an AI recommendation about a news story. A recent study about AI aversion found that implicit prejudice about AI, which is ephemeral and diminishes when people get familiar with AI [102]. Thus, users will treat AI recommendations as endorsements, similar to expert recommendations, and are more likely to rely on experiential cognition and less likely to invoke rational cognition. Thus:

*Hypothesis 2 (H2): Labeling a news recommendation as coming from AI will induce users to rely more on fast cognition, compared with no recommendation.*

### **News Sharing**

News sharing is a common behavior of social media users [14, 15] and serves the need for self-expression [47]. Although sharing a story is often done with minimal effort (a quick click), individuals can still feel like they get the benefits on social media because the act of sharing a story is more about supporting a friend or ingroup than about the content of the specific news story [50]. This does not require the related story to be true, as gestures of loyalty are valued regardless of the circumstances [85]. Users can choose to invoke rational System 2 cognition, allowing some experiential cognition results to be translated into actions while suppressing others [76]. When System 2 rational cognition dominates, individuals spend more time thinking about the incoming information and are also more cautious about subsequent decisions [45]. Deliberation by System 2 cognition can prevent users from believing fake news [5], such that it reduces the likelihood that users share fake news.

However, most users are not willing to expend the effort needed to use System 2 cognition [65, 77] due to the hedonic nature of social media, the information volume on social media, and the human nature of saving cognitive resources. This means that it is not uncommon for experiential cognition to guide actions such as liking and reposting stories [97]. As users become familiar with social media, news sharing can become so automatic that making this decision no longer requires effortful, rational thinking. A study of Twitter data shows that most people who share stories *do not* read them before they share them [31], providing further support for the idea that social media sharing is more heavily driven by experiential cognition than rational cognition. Therefore, we hypothesize:

*Hypothesis 3 (H3): The more social media users rely on fast cognition, the more likely they are to share news stories.*

*Hypothesis 4 (H4): Fast cognition mediates the relationship between recommendation and fake news sharing.*

## News Veracity

Fake news can be subtle or sensational, presenting novel impressions of what is happening [28]. Such stories can be attention-grabbing [85], particularly if a reader is not thinking deeply about the headline. Fake news stories are often designed to trigger confirmation bias by presenting a story that aligns with a user's prior beliefs [37, 75] such that experiential System 1 cognition quickly produces a feeling that the story is true. This fast cognition reduces their ability to discern fake news from true news [65, 81], which influences their news sharing behavior [5]. Prior studies on fake news found that System 2 cognition reduces users' belief of fake news but does not affect the belief of true news stories [5]. Rational, thoughtful engagement with fake stories enables individuals to spot fallacies and uncover deceit [5]. Rational System 2 cognition is better able to help users detect fake news by prompting them to think critically about what they are seeing, rather than reacting instinctively [81]. In contrast, fast cognition leads them to make reasoning errors so they are less likely to be able to detect the subtle cues that identify a story as fake [81]. As a result, they are more likely to believe fake news stories are true when they use fast cognition [5, 81].

Past research has used the Cognition Reflection Test to assess a person's overall tendency to use fast cognition (i.e., rely on System 1 cognition) and found that individuals with a greater tendency to use fast cognition were more likely to believe fake news stories [81]. Belief that a news story is true is an important factor in the decision to share it [5, 81]. Social media users are more likely to share a story they believe to be true than a story they believe to be false [48, 49, 67]. These findings suggest that reducing the use of fast cognition may reduce the sharing of fake news stories.

In summary, the use of System 2 cognition is important in identifying fake news stories and social media users are less likely to share stories they believe to be fake. Therefore, the use of System 2 cognition is more likely to reduce the sharing of fake stories while having less effect on the sharing of true stories. Thus, news veracity (i.e., fake news versus true news) will moderate the relationship between cognition and news engagement. Specifically:

*Hypothesis 5 (H5): The positive effect of reliance on fast cognition on news sharing will be stronger for fake news stories than for true news stories.*

## Overview of the Studies

In general, there are two forms of AI commonly deployed on the Internet: visible agents and embedded algorithms [35]. Visible agents are common, with chatbots and other conversational agents being ubiquitous [89]. Likewise, embedded algorithms are common on social media and a host of other media contexts (e.g., video recommendations on Netflix). We conducted three studies examining both forms of AI (see Table 2).

Study 1 examined visible agentic AI with a human likeness (e.g., a name, first-person pronouns), as this increases the acceptance of AI [89] and is becoming increasingly popular due to its benefits over machinelike AI [9, 90]. Studies 2 and 3 examine the influence of AI presented as an embedded algorithm. We theorize that social influence will be stronger for the agentic AI in Study 1 than for the embedded AI in Studies 2 and 3, because the agentic AI has a more human-like appearance. Studies 1 and 2 present AI in a positive light because this is how social media presents AI (e.g., <https://www.facebook.com/help/1257205004624246>); Study 3 changes this to present AI with a neutral framing.

**Table 2.** Overview of studies.

|                  | Study 1                                                                                       | Study 2                                                            | Study 3                                                                         |
|------------------|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------|---------------------------------------------------------------------------------|
| AI design        | AI Agent (human-like)                                                                         | Embedded AI (algorithmic)                                          | Embedded AI (algorithmic)                                                       |
| Social Influence | Strong                                                                                        | Moderate                                                           | Moderate                                                                        |
| AI Framing       | Positive                                                                                      | Positive                                                           | Neutral                                                                         |
| Purpose          | Examine the effect of AI recommendations on cognition and behavior (as well as human experts) | Examine the effect of AI recommendations on cognition and behavior | Examine the effect of AI recommendations on cognition and behavior in GenAI era |

Abbreviation: AI, artificial intelligence.

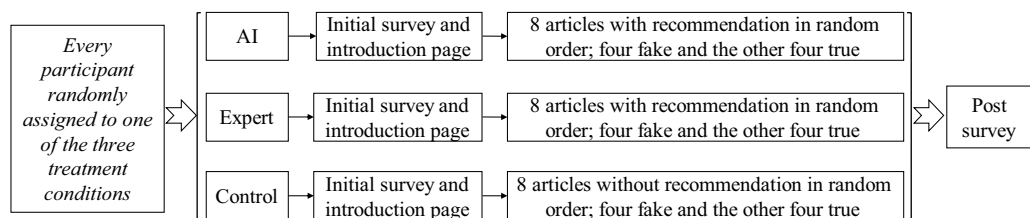
## Study 1: AI as a Visible Agent

### Participants

We collected data from 360 US participants from Cloud Research who were paid \$2 in April 2022. Research shows that experimental data from such participants is equivalent in quality to data from other commonly used sources such as students and tends to be more representative of the population as a whole [99] and existing fake news studies have shown the appropriateness of using these platforms to collect data [49, 65, 77, 81]. After removing the cases failing the attention check questions, we have 347 valid responses. About 44% were female. About 41% had a bachelor's degree. About 32% self-identified as Conservative (extremely conservative, moderately conservative, and slightly conservative), 51% as Liberal (extremely liberal, moderately liberal, and slightly liberal), and 17% as independent.

### Treatments

We used a between- and within-subject design with repeated measures (Figure 2). To avoid any confusion about the recommender, each participant saw only one recommender (or no recommender in the control treatment); this also enabled us to measure cognition once, rather than after each story, which would have been intrusive and enabled them to guess the research question and change their behavior. Participants were randomly assigned to one of the three treatments (AI, expert, or no recommender). We designed the AI recommender to resemble the AI agents currently used on Facebook and the numerous websites and applications that provide customer services (e.g., [58]). As a manipulation check, participants in the three treatment groups were asked to identify who recommended news to them; no participant provided an incorrect answer. As a randomization check, we assessed the demographics of participants assigned to the different treatments, finding no significant

**Figure 2.** Design of Study 1.

differences between the treatments (gender:  $p = 0.689$ ; age:  $p = 0.248$ ; political orientation:  $p = 0.976$ ; Facebook usage:  $p = 0.932$ ); we conclude that the randomization was successful.

News veracity is a within-subject factor, referring to the actual truthfulness of a news story, which is a binary concept in this paper. Half the stories were verified as false and half were verified as true. We coded news veracity at the story level, assigning fake stories a value of 1 and true stories a value of 0. In the pretest, for each headline, we asked participants a direct question about the perceived accuracy, and fake news headlines were rated as less accurate than true news headlines (see the headlines in Online Supplemental Appendix 2.3, Table 2.4).

## **Task**

Participants read an introduction about the task and the treatment they were assigned to (those in the control group only saw task information; see Online Supplemental Appendix 2.1). They then viewed all eight news headlines and reported their intentions to share each headline, one at a time in random order. The headlines were relevant at the time of the study and were drawn from a preliminary study that assessed their political leaning and emotionality (see Online Supplemental Appendix 1). The study ended with participants answering questions about cognition and demographics.

## **Measurements**

### **Dependent Variables**

The dependent variable is users' intention to share each headline [65]. Intentions are reliable indicators of actual behavior [4, 68] because few people accidentally share stories (and our research is interested in the intentional sharing, not the accidental sharing). This is why intentions are widely used to investigate the sharing of fake news [39, 48]. Sharing intentions of crowd workers are strongly related to their actual popularity (i.e., retweets) of a news story on social media platforms [68]. See Online Supplemental Appendix 3 for all items.

### **Mediator**

Fast cognition, the extent to which participants relied more on experiential System 1 than on rational System 2 cognition, was measured using six items adapted from Novak and Hoffman [72]. Three items measured reliance on experiential System 1 cognition, and three measured the reliance on rational System 2 cognition (see Online Supplemental Appendix 3). Fast cognition is the extent to which a user relies more on experiential cognition than on rational cognition [81]; as a result, we created a variable that was the difference between the two (i.e., experiential System 1 cognition minus rational System 2 cognition, following the practice of Martelet al. [64] who used the difference between the two in their experiments 2-4). Thus, the more a participant relies on experiential cognition versus rational cognition, the larger the value of this fast cognition measure. This measurement approach can better capture an individual's cognition in a specific situation (i.e., news on social media) compared with the Cognitive Reflection Test, which assesses propensity to use System 1 cognition as a general personality trait. The statistical descriptions of the constructs are in Online Supplemental Appendix 3.

As a validation check of self-reported cognition measures, we also recorded the time participants spent on each headline. Time is an objective indicator of rational cognitive effort and is widely used in investigating cognition because the use of System 2 cognition requires more time than the use of System 1 cognition [45]. The mean reading time for each headline was about 18 seconds, which was significantly positively correlated with self-reported rational cognition ( $b = 0.075$ ;  $p = 0.013$ ). We conclude that the approach of Novak and Hoffman [72] was appropriate in our context. We also note that these times match with studies of real user behavior, which found that most stories are shared on social media with less than 15 seconds of consideration [65] and frequently without reading the article [107].

### **Control Variables**

Confirmation bias is a major factor influencing users' engagement with news stories [67], so we included it as a control variable. Political orientation was measured on a seven-point scale with 1 indicating extremely liberal, 4 independent, and 7 extremely conservative [66]. To avoid any desirability bias effect and potential influence on users' cognition, we measured confirmation bias using the implicit method of Pennycook and colleagues [78, 80] by multiplying the political leaning of a headline and the political affiliation reported by each participant (rescaled to -3 to 3, 0 = independent). We also included demographic variables (e.g., gender, age, Facebook usage frequency).

### **Results**

The measurement model displayed satisfactory convergent validity, discriminant validity, and reliability (see Online Supplemental Appendix 3). Our theoretical model argues that fast cognition mediates the effects of recommendations on news sharing. There are several approaches to testing moderated mediation. We used the three-step method of Baron and Kenny [7] because the Hayes PROCESS and its MLMED extension cannot be used with multilevel data like ours with the mediator at level 2 and multiple outcomes at level 1 [38].

The first step is to validate the relationship from the independent variables to the outcome variable (without the theorized mediating variable). The second step is to show that the independent variables influence the mediating variable. The third and last step is examining the simultaneous impacts of the independent variables and the mediating variable on the outcome variable to assess whether there is a full or partial mediation; this also needs to include interactions between the mediator and moderator, as the effect of the mediator may differ by the moderating condition [38]. For Steps 1 and 3, we have data at the participant level and multiple observations for each participant at the news story level, so we used multilevel mixed models in STATA with random intercepts and robust standard errors, and the control group was treated as the baseline [18, 65]. Step 2 used a linear regression in STATA to analyze the relationship between the recommendation and the participant's fast cognition because these data are all at level 2 (participant level).

#### **Step 1: The Overall Effects of News Recommendation**

The first step assesses whether the independent variables have any overall effect on the outcome variable, without considering the mediator. The results show that both types of

**Table 3.** Study 1 statistical results.

|                       | Model 1<br>News Sharing | Model 2<br>Fast Cognition | Model 3<br>News Sharing |
|-----------------------|-------------------------|---------------------------|-------------------------|
| Independent Variables |                         |                           |                         |
| Fake                  | -0.388***               |                           | -0.363***               |
| AI                    | 0.443*                  | 0.524***                  | 0.334                   |
| AI X Fake             | 0.255*                  |                           | 0.224                   |
| Expert                | 0.472*                  | 0.630***                  | 0.345                   |
| Expert X Fake         | 0.258*                  |                           | 0.215                   |
| Mediators             |                         |                           |                         |
| Fast Cognition        |                         |                           | 0.185*                  |
| Fast Cognition X Fake |                         |                           | 0.104*                  |
| Controls              |                         |                           |                         |
| Confirmation bias     | 0.454***                |                           | 0.451***                |
| Conservative          | 0.071                   | 0.332                     | -0.003                  |
| AI X Conservative     | -1.020**                | -0.667*                   | -0.862*                 |
| Expert X Conservative | -0.736                  | -0.605*                   | -0.593                  |
| Female                | 0.055                   | 0.173                     | 0.014                   |
| Facebook              | 0.281                   | -0.001                    | 0.281                   |
| Younger Adults        | 0.994*                  | 0.159                     | 0.956*                  |
| Older Adults          | 0.354                   | -0.486                    | 0.468                   |
| Constant              | 2.426***                | -0.413***                 | 2.510***                |
| Model Details         |                         |                           |                         |
| N (Participants)      | 347                     | 347                       | 347                     |
| N (Observations)      | 2776                    |                           | 2776                    |

Note: \* $p < 0.05$ . \*\* $p < 0.01$ . \*\*\* $p < 0.001$ . Abbreviation: AI, artificial intelligence.

news recommendations had significant effects on sharing, with both main and interaction effects (see Model 1 in Table 3). AI and expert recommendations *increased* both true and fake news sharing, compared with no recommendation, and this was stronger for fake news stories.

### Step 2: The Effects of Recommendation on Cognition

The second step was to assess whether different types of news recommendations influenced the theorized mediator (i.e., fast cognition, the relative reliance on experiential rather than rational cognition). This analysis omits confirmation bias and whether the headline is fake or true because both are tied to specific headlines and this analysis is at the participant level, not at the headline level. AI and expert recommendations had significant effects on fast cognition (see Table 3, Model 2). Compared to the control group, participants receiving a story labeled as recommended by AI or an expert relied on their fast cognition to a greater extent. Therefore, H1 and H2 were supported.

### Step 3: The Full Model

The third step was to examine whether the mediator affects the outcome variable and whether the effects of the independent variables still exist once the moderated mediation is included. We performed another multilevel mixed-effect regression in Stata, including the mediator (see Model 3 in Table 3). The analysis shows that fast cognition significantly increased news sharing. Thus, H3 was supported. After including fast cognition in the regression, the direct effects of AI and expert recommendations on news sharing and the

interactions all disappeared. Thus, fast cognition fully mediates the effects of news recommendations on news sharing. H4 is supported.

As a post-hoc test, we used two Wald tests to compare the coefficients of AI with those of human experts on fast cognition ( $p = 0.724$ ), and news sharing ( $p = 0.954$ ). We found no significant differences. Thus, there is no evidence to suggest that social media users treated an AI recommendation any differently than a human expert recommendation.

Model 3 also shows that the moderating effect of news veracity (i.e., true or fake news) between cognition and news sharing was significant. Fast cognition positively influenced sharing, and the interaction between cognition and news veracity was positive and significant, which indicates that this effect was stronger for fake news stories, compared with true news stories. Thus, H5 was supported. The patterns found in Study 1 were consistent with our hypotheses that: 1) news recommendations increase news sharing; 2) fast cognition fully mediates the relationship between news recommendation and news sharing; and 3) the relationship between fast cognition and news sharing was moderated by news veracity.

## Study 2: AI as an Embedded Algorithm

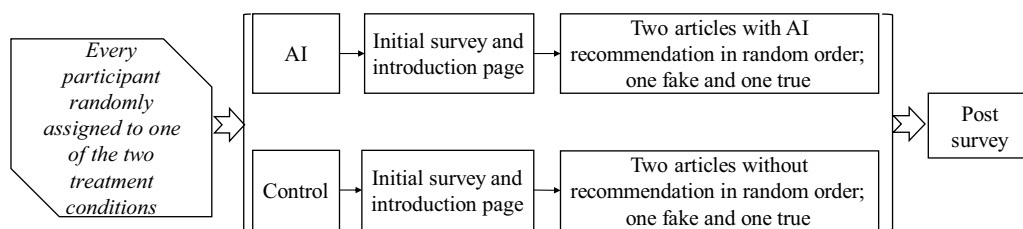
Firms often deploy AI in the form of an embedded algorithm and much research has examined this form of AI [43, 65, 111]. Therefore, in this experiment, we examine the effects of a different form of AI (a more machinelike embedded algorithm) by following the common labeling designs used by major social media platforms. The display of embedded AI on social media platforms is simple, straightforward, and framed professionally and positively [27, 59]. Social media platforms provide explanations about their AI systems by emphasizing the benefits AI can bring to users<sup>1</sup> and, therefore, we used this approach to describe AI.

We propose that the recommendation from a more machinelike AI algorithm can affect users' cognition and induce them to share news in a manner similar to more humanlike visible AI agents. The social influence effects of embedded AI are likely to be weaker than those of agentic AI because the agentic AI appears in a more humanlike form and thus is more like a human expert than the more machinelike embedded AI. Nonetheless, we hypothesize that embedded AI will have social influence effects on user cognition.

Generative AI has entered the public consciousness, and most people are now more cognizant of AI and its capabilities. As a result, it is important to understand whether the reactions of users to AI have changed (Study 1 was conducted before the widespread use of ChatGPT, and Study 2 was conducted after). Thus, we conducted Study 2 to reconfirm our findings and extend Study 1 to the embedded form of AI. Since the effects of AI recommendations are the locus of interest, expert recommendations are not considered in Study 2. Thus, H2, H3, H4, and H5 will be reexamined.

## Participants

We recruited 400 U.S. participants from Cloud Research in May 2024 who were paid \$1. After removing the 13 participants who failed the attention check questions, we obtained 387 valid responses. About 45% were female. About 49% had a bachelor's degree. About 48% self-identified as Conservative, and 52% as Liberal.



**Figure 3.** Design of Study 2 and 3.

### Treatments

We used a between- and within-subject design with repeated measures at the individual level (Figure 3). Participants were randomly assigned to either the AI recommendation treatment or the control treatment (no recommendation) and were shown the two headlines (one true, one false). We designed the more machinelike AI algorithm recommendation to look like the AI labels currently used on major social media platforms (e.g., Facebook, Instagram)<sup>2</sup> and attached the label to both headlines. The label was direct and straightforward, with the phrase “Suggested by AI.” The design was adapted from the Facebook website, stressing the rationale and benefits. As a manipulation check, participants in the treatment group answered a question about the source of news recommendation; only after answering this question correctly, participants could proceed to the news reading task section. As a randomization check, we assessed the demographics of participants assigned to the two groups, and found no significant differences between the treatment group and the control group (gender:  $p = 0.386$ ; age:  $p = 0.261$ ; political orientation:  $p = 0.951$ ; Facebook usage:  $p = 0.692$ ). Thus, the randomization was successful.

### Task

Participants read an introduction about the task. They then viewed two news headlines and reported their intentions to share each headline, one at a time in random order. The headlines were relevant at the time of the study, one fake and one true. These two headlines were selected based on the political leaning of the participants, ensuring that Democrats saw left-leaning headlines while Republicans saw right-leaning headlines. Participants then answered questions about cognition. All the headlines are selected after a new pretest to make sure that the political leaning of each headline aligns with our anticipation and the perceived accuracy is on a similar level (see the headlines in Online Supplemental Appendix 2.3, Table 2.5).

### Measurements

All the construct measurements are identical to Study 1. As a validation check of self-reported cognition measures, we also recorded the time participants spent on the reading task. The mean reading time for each headline was about 24 seconds, which was significantly positively correlated to self-reported rational cognition ( $b = 0.092$ ;  $p = 0.009$ ), as in Study 1.

**Table 4.** Study 2 statistical results.

|                       | Model 1<br>News Sharing | Model 2<br>Fast Cognition | Model 3<br>News Sharing |
|-----------------------|-------------------------|---------------------------|-------------------------|
| Independent Variables |                         |                           |                         |
| Fake                  | -0.342***               |                           | -0.350***               |
| AI                    | -0.136                  | 0.336*                    | -0.206                  |
| AI X Fake             | 0.388**                 |                           | 0.400**                 |
| Mediators             |                         |                           |                         |
| Fast Cognition        |                         |                           | 0.236*                  |
| Fast Cognition X Fake |                         |                           | -0.089                  |
| Controls              |                         |                           |                         |
| Confirmation bias     | 0.289**                 |                           | 0.284**                 |
| Conservative          | 1.262***                | 0.491***                  | 1.167***                |
| AI X Conservative     | -0.997**                | -0.478*                   | -0.905*                 |
| Female                | -0.211                  | 0.134                     | -0.236                  |
| Facebook              | 0.386***                | -0.035                    | 0.392***                |
| Younger Adults        | 0.789*                  | 0.320                     | 0.728*                  |
| Older Adults          | -0.562                  | -0.446                    | -0.475                  |
| Constant              | 2.471***                | -0.384***                 | 2.548***                |
| Model Details         |                         |                           |                         |
| N (Participants)      | 387                     | 387                       | 387                     |
| N (Observations)      | 774                     |                           | 774                     |

Note: \* $p < 0.05$ . \*\* $p < 0.01$ . \*\*\* $p < 0.001$ . Abbreviation: AI, artificial intelligence.

## Results

As in Study 1, we used STATA to analyze the data following the 3-step method of Baron and Kenny [7] (Table 4). Model 1 indicates that AI recommendations increase the sharing of fake news articles. Model 2 shows that AI recommendation was positively related to fast cognition, which supports H2. Model 3 shows that fast cognition was a positive predictor of the sharing intention. Thus, H3 was supported again. However, the effect of fast cognition on news sharing was not stronger for fake news, as the interaction between cognition and news veracity was not significant; as a result, H5 was not supported. The main effect of AI on fake news sharing remained significant even after accounting for fast cognition, suggesting a partial mediation of AI recommendations on fake news sharing. H4 is supported.

If we compare the results of Study 2 with those of Study 1, we observe that the main findings are consistent. Adding an AI label to a fake news story increases the likelihood that it will be shared (as shown by the AI x Fake interaction in Model 1 in both studies). This is because the AI label increases the use of fast cognition (as demonstrated by the main effect of AI in Model 2 in both studies). This increase in fast cognition increases the likelihood of sharing (as shown by the main effect of fast cognition in Model 3 in both studies).

We theorized that there might be differences between the studies because the agentic AI in Study 1 is likely to have greater social influence than the embedded AI in Study 2. The results support this: the agentic AI in Study 1 had a stronger effect; the main effect of AI is significant in Study 1 Model 1, indicating AI labels affect both true and fake articles, whereas it is not significant in Study 2 Model 1, indicating that AI labels have no effect for true articles). Study 1 also shows that fast cognition has a stronger effect for fake articles than true articles (the significant fast cognition x fake interaction in Model 3), but this is not significant in Study 2, suggesting a weaker effect. In Study 1, fast cognition fully mediates the effects of AI labels, whereas it only partially mediates the effects of AI labels in Study 2 (as there remains a significant AI by Fake interaction in Model 3). Thus, in a GenAI world where users have come to expect AI to be very capable [51] (i.e., Study 2), users may rely

more on AI labels without thinking [51], and, therefore, cognition has only a partial mediation role.

### Study 3: AI in the Post-GenAI Era

Generative AI has shaped many aspects of life and has become a regular tool in everyday life. People have become increasingly familiar with the term “AI,” regardless of their actual understanding of it. As a result, it is important to understand how users see AI in the post-GAI era (Study 1 was conducted before the widespread use of ChatGPT, and Study 2 was conducted during its rise). Thus, Study 3 was conducted to reconfirm our findings and extend Study 2. We also revised the description of AI to be more neutral, to better understand the effects if social media platforms are not permitted to promote the positive aspects of AI. The design of Study 3 is similar to Study 2, and H2, H3, H4, and H5 were reexamined.

#### Participants

We recruited 400 U.S. participants from Prolific in May 2025, one year after Study 2. Every participant received \$1.50 as compensation. After removing the 40 participants who failed the two attention check questions (e.g., “please select strongly agree if you are reading”), we obtained 360 valid responses. About 54% were female. About 38% had a bachelor’s degree. About 55% self-identified as Conservative and 45% as Liberal.

#### Experiment Design

As in Study 2, we used a between- and within-subject design with repeated measures at the individual level, counterbalanced labels, and random assignment to treatment. Participants were shown the four headlines (two true, two false). We selected up-to-date news headlines through a pretest to ensure all headlines were relevant and not greatly different in the perceived accuracy and length (see the headlines in Online Supplemental Appendix 2, Table 2.6). Participants were shown headlines that aligned with their political preferences.

Most elements of the experiment’s design were the same as Study 2, including the AI label, the procedure, and the measurements of the dependent variable and cognition. Four changes were made compared to Study 2. First, to better capture the sharing behavior of participants, we provided a clickable share button after each headline, similar to those found on most social media platforms. If they clicked it, we displayed a message saying that a shareable link to this story would be provided at the end of this study. Doing this enabled us to get more directly at users’ behaviors, rather than their intentions. Second, in Study 2, we informed participants that the platform was Facebook, whereas in Study 3, we avoided naming any specific platform. Third, we framed AI in a neutral way, rather than the positive way it was framed in Studies 2 and 3 (see Online Supplemental Appendix 2). Finally, we updated the headlines, as some headlines used in Study 2 had become outdated by the time of Study 3 (e.g., referring to President Biden); see Online Supplemental Appendix 2.

As a randomization check, we assessed the demographics of participants assigned to the two groups (AI VS control) and found no significant differences between the treatment group and the control group (gender:  $p = 0.142$ ; age:  $p = 0.876$ ; political orientation:  $p = 0.841$ ; Facebook usage:  $p = 0.379$ ). Thus, the randomization was successful. The mean time

spent on each headline is about 39 seconds (slightly longer than in Studies 1 and 2, likely because this includes time spent thinking about clicking on the share button and reading the message we showed after clicking), which was again positively and significantly related to self-reported rational cognition, as in Study 1 and Study 2 ( $b = 0.120$ ,  $p = 0.023$ ).

## Results

As in Study 1, we used STATA to analyze the data following the 3-step method of Baron and Kenny [7] (Table 5). Model 1 shows that AI increases the sharing of fake news articles. Model 2 shows that AI was positively related to fast cognition, and Model 3 shows that fast cognition was a positive predictor of the sharing intention. Fast cognition partially mediated the effect of AI recommendations, so H4 was supported. The findings align with those of Study 2, as H2, H3, and H4 were supported, H5 was not supported.

Figure 4 presents the sharing results graphically. Without disclosing that AI is making recommendations, true news is shared more than fake news. When the role of AI is disclosed, the pattern reverses and people are more likely to share fake news than true news. This shows that social media users view AI as a useful tool in curating news and are willing to rely on it.

Figure 5 shows how AI recommendations increase fast cognition (on a scale of -3 to +3). When an AI recommendation is present, the reliance on fast cognition increases.

If we compare the results of Study 3 with those of Study 2, we see that the main results are the same. Adding an AI label to a fake news story increases the likelihood that it will be shared (as shown by the AI x Fake interaction in Model 1 in both studies). This is because the AI label increases the use of fast cognition (as shown by the main effect of AI in Model 2 in both studies). This increase in fast cognition increases the likelihood of sharing (as shown by the main effect of fast cognition in Model 3 in both studies). We note that the fake news headlines in this study were

**Table 5.** Study 3 statistical results.

|                       | Model 1<br>News Sharing | Model 2<br>Fast Cognition | Model 3<br>News Sharing |
|-----------------------|-------------------------|---------------------------|-------------------------|
| Independent Variables |                         |                           |                         |
| Fake                  | -0.148                  |                           | -0.137                  |
| AI                    | -0.345                  | 0.319*                    | -0.415                  |
| AI X Fake             | 0.295*                  |                           | 0.275*                  |
| Mediators             |                         |                           |                         |
| Fast Cognition        |                         |                           | 0.185*                  |
| Fast Cognition X Fake |                         |                           | -0.089                  |
| Controls              |                         |                           |                         |
| Confirmation bias     | 0.297***                |                           | 0.298***                |
| Conservative          | -0.165                  | 0.434**                   | -0.273                  |
| AI X Conservative     | 0.417                   | -0.323                    | 0.497                   |
| Female                | 0.314                   | 0.151                     | 0.277                   |
| Facebook              | 0.266**                 | -0.080                    | 0.286**                 |
| Younger Adults        | -0.447                  | -0.203                    | -0.396                  |
| Older Adults          | 0.068                   | -0.047                    | 0.080                   |
| Constant              | 4.583***                | -0.361**                  | 4.667***                |
| Model Details         |                         |                           |                         |
| N (Participants)      | 360                     | 360                       | 360                     |
| N (Observations)      | 1440                    |                           | 1440                    |

Note: \* $p < 0.05$ . \*\* $p < 0.01$ . \*\*\* $p < 0.001$ . Abbreviation: AI, artificial intelligence.

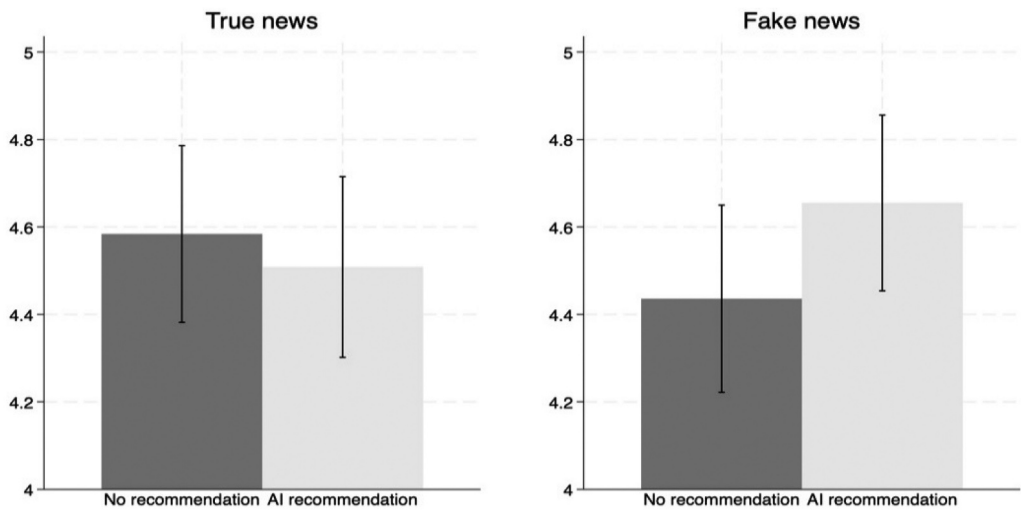


Figure 4. Recommendations and news sharing.

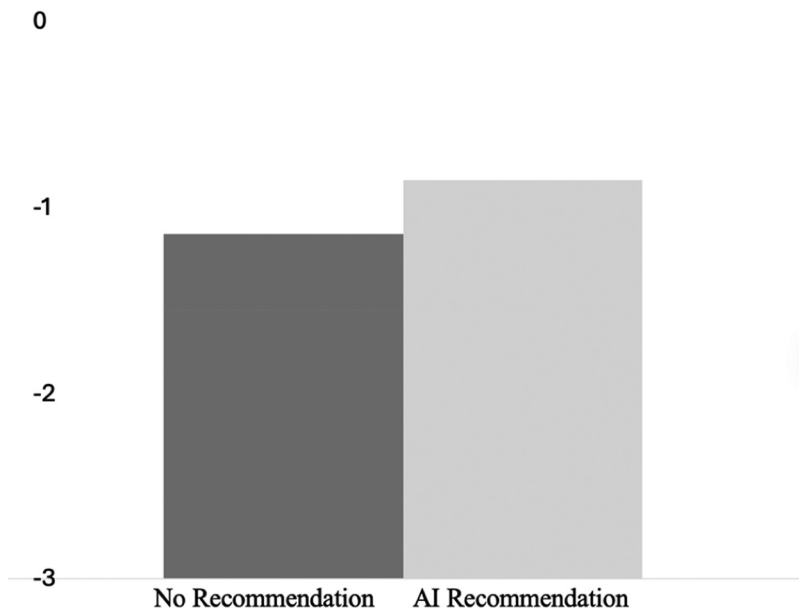


Figure 5. Recommendations and fast cognition.

more challenging for participants to detect (Table 4). We captured whether participants clicked on the share button for each story and used a multilevel logistic regression to test the relationship between the button clicking behavior and sharing intention: sharing intention predicted clicking the share button ( $b = 0.380$ ,  $p < 0.001$ ).

General Discussion

AI plays a central role in recommending news stories displayed on social media, but most users remain unaware of this [cf. 65, 132]. This lack of transparency about AI [35, 87] has prompted policy experts to suggest that social media platforms should clearly label content recommended by AI as a way to combat fake news sharing. Policymakers believe that labeling stories as recommended by AI will serve as a warning, encouraging users to think more carefully about the news stories they see, and thereby reducing the likelihood of sharing fake news stories promoted by AI. In other words, policymakers assume that social media users will exhibit a bias against AI [102] and see AI labels as a warning.

Our research shows that such a policy change to increase transparency about AI use is likely to backfire and *increase* fake news sharing. Our findings reveal that labeling a story as recommended by AI had similar effects on users’ cognition and news sharing behavior as labeling it as recommended by human experts. Users were *more likely* to share fake news when those stories were explicitly labeled as recommended by AI. We used the lens of social influence theory to argue that users would treat an AI recommendation label as a social cue and thus—just like human expert recommendations—such labels *reduce* the likelihood that users will engage in effortful rational cognition rather than quick experimental cognition.

Our results generally support our theorizing (Table 6). Labeling stories as recommended by AI increased the likelihood that users would share them. When encountering an AI recommendation, users tended to increase their reliance on experiential System 1 cognition compared with rational System 2 cognition, just as they would if the story was labelled as recommended by a human expert. Past research has demonstrated that reliance on experiential thinking increases belief in fake news, while increased rational thinking is beneficial for recognizing fake news [10, 67, 81]. Our results move beyond this prior research by showing that 1) AI recommendation labels influence cognition; 2) cognition affects sharing; and 3) cognition may have stronger effects for true news.

We also found that conservatives react differently to AI recommendations compared to liberals and moderates: they were less influenced by AI recommendations. This is likely due in part to the trend that conservatives are more skeptical towards AI and its deployment in

Table 6. Summary of results.

| Study 1<br>Humanlike AI                  |                                               | Study 2<br>Embedded AI                             | Study 3<br>Embedded AI                             |
|------------------------------------------|-----------------------------------------------|----------------------------------------------------|----------------------------------------------------|
| AI Effects without considering Cognition |                                               |                                                    |                                                    |
| AI                                       | AI Increases Sharing of all News              | No Main Effect                                     | No Main Effect                                     |
| AI X Fake                                | AI increases Sharing of Fake News             | AI increases Sharing of Fake News                  | AI increases Sharing of Fake News                  |
| AI Effects on Cognition                  |                                               |                                                    |                                                    |
| Fast Cognition                           | AI increases Use of Fast Cognition            | AI increases Use of Fast Cognition                 | AI increases Use of Fast Cognition                 |
| AI Effects considering Cognition         |                                               |                                                    |                                                    |
| Fast Cognition                           | Fast Cognition Increases sharing of all News  | Fast Cognition Increases sharing of all News       | Fast Cognition Increases sharing of all News       |
| Fast Cognition X Fake                    | Fast Cognition Increases sharing of Fake News | No Additional Effect for Fake News                 | No Additional Effect for Fake News                 |
| AI                                       | Effect explained by Cognition                 | No Main Effect                                     | No Main Effect                                     |
| AI X Fake                                | Effect explained by Cognition                 | Additional Effect not fully explained by Cognition | Additional Effect not fully explained by Cognition |

Abbreviation: AI, artificial intelligence.

general [110, 113]. Furthermore, this is aligned with the current finding that conservatives seek control, while liberals seek autonomy and thus favor using AI technology [16].

This study suffers from the usual limitations of experimental research. We studied participants from one country reading artificial headlines to provide control. Future research should examine the effects of AI and human expert recommendations in field settings in different countries. We followed the approach of Novak and Hoffman [72] and used self-reports of cognition. People tend to overestimate their reliance on rational System 2 cognition and underestimate their reliance on experiential System 1 cognition [45], so self-reports tend to be a conservative estimate—when self-reports find increased reliance on System 1 and decreased reliance on System 2, the actual differences tend to be much larger [45].

### **Implications for Future Theory and Research**

This study extends past research on fake news to show that labeling news stories as recommended by AI *increases* fake news sharing. Thus, being more transparent about the role of AI is likely to *increase* the sharing of fake news (and likely true news as well). The effects of AI recommendation have been investigated in many fields such as media (e.g., Netflix videos), marketing (e.g., Amazon products), and medical decision-making (e.g., medical treatments), and the results are diverse and context-dependent [11, 21], finding either aversion [11, 20] or appreciation [57]. Early research showed that algorithm aversion was due to errors by AI, but more recent research shows that algorithm aversion is much broader than originally conceived and can occur without the AI making errors [102].

We find that in the context of social media news, people treat AI in a similar manner to human experts: an AI recommendation triggers fast cognition and a general acceptance of the story. This effect is stronger when the AI is depicted in a more humanlike way, indicating that AI has social influence not only when it is presented in a form similar to human conversations as argued by CASA [69, 70]. In other words, it is the AI itself that has social influence, not just the human-like social setting that triggers influence as theorized by CASA.

Algorithm appreciation may be dominant in this situation because social media users see AI as a reliable tool with less bias than humans and fewer errors [42]. It may also be that users are used to accepting AI recommendations on other media platforms (e.g., Netflix) so they transfer this behavior to social media platforms. User factors may also influence the effects of AI recommendations, such as economic status, cultural background, and digital literacy. Research is needed to clarify the boundary conditions of the effects.

Second, we identified the theoretical mechanism that explains why AI disclosure increases sharing: fast cognition. Past research suggests that a lack of rational System 2 cognition is a primary explanation for erroneous fake news beliefs [10, 67, 81]. Our results show that labeling news stories as recommended by AI increases the reliance on System 1 fast cognition compared with System 2 cognition, which is the opposite of what policymakers believe is likely to happen. We used a self-reported measure of cognition, extending previous research that used trait-level measures such as the Cognitive Reflection Test [64]. Future studies should delve more deeply into AI effects on cognition, because there are many potential problems if people routinely rely on AI and avoid thinking for themselves. Future studies can

investigate the specific reasons why human cognition is influenced by AI. NeuroIS methods can be leveraged to measure human cognition, such as electroencephalogram (EEG) [67].

Third, although algorithm aversion has been an issue in many contexts [20, 86, 102], we found no evidence of algorithm aversion for AI recommendations of fake news stories. This may be because of the hedonic nature of social media as well as the lack of immediate consequences for fake news engagement. More research is needed to understand how users react to AI recommendations after seeing the AI recommend fake stories. We used two versions of AI design in this study and the degree to which social media users are influenced is not equal. We suspect that the reason is related to the human like nature of the agentic AI. More studies can compare more designs and investigate the underlying reasons.

### **Implications for Practice**

First and foremost, our study indicates that labeling social media news stories as recommended by AI is likely to *increase*—not decrease—the spread of fake news. AI recommendation labels trigger users to rely on System 1 cognition, which increases the likelihood that they will share fake news. Thus, as a policy intervention to reduce fake news, increasing transparency by labeling stories as recommended by AI is likely to backfire.

Transparency about the role of AI has become a public debate [87, 109], prompted by users' general lack of awareness [52, 108]. Many platforms have been reluctant to embrace transparency, fearing it might reduce user engagement. Yet, many of the purported benefits of transparency failed to materialize in our study because policymakers misunderstand the implications of social influence theory [19], and fail to see that an AI recommendation is a social cue that *reduces*—not increases—the likelihood of adopting effortful cognition. Users have come to rely on AI to provide good recommendations of music, videos, and other media [24, 46, 56] and, as a result, see AI labels as a signal of quality, not as a warning.

Therefore, due to these results, policymakers (especially those in Western countries like the American context we studied) seeking to combat fake news should be cautious in attempting to use AI labels to reduce the spread of fake news. Our research suggests that placing AI “warnings” on stories *increases* the spread of fake news because it increases the use of fast cognition [65, 81]. Therefore, a better strategy is to nudge social media users away from fast cognition toward more effortful System 2 cognition when making the decision to share or not share [65, 81]. For example, something as simple as an accuracy prompt may reduce fake news sharing (e.g., suggesting that users pause to think if the story is accurate before they share) [66, 78, 82]. Other forms of AI labels can only be deployed on a large scale after testing and ensuring their effectiveness in reducing fake news sharing.

Our results also have implications for marketers and advertisers, as well as governments and non-profits striving to spread public service announcements. Organizations often invest money and effort to take advantage of electronic word-of-mouth from users to promote their products, services, or messages. Our study shows that if organizations want to increase the consumption (i.e., reading) and/or spread of their advertisements or public service announcements on social media platforms, an automated promotion strategy using AI is likely to be no less effective than leveraging human word-of-mouth on social media.

## Conclusion

Our study shows that labeling social media stories as recommended by AI is likely to backfire and increase the spread of fake news stories. An AI label had the same effect as labeling a story as recommended by an expert; both increased the likelihood of sharing fake news because the labels induced users to rely more on experiential System 1 cognition than rational System 2 cognition. Furthermore, these effects were not materially affected by the design of AI: social media users reacted to a more machinelike embedded AI algorithm and a visible AI agent in similar ways. This indicates that social influence from a recommender can be an important factor in driving the spread of fake news and AI agents are no less influential than humans in shaping users' news engagement behavior.

## Notes

1. For example, see <https://about.fb.com/news/2021/03/more-control-and-context-in-news-feed/>
2. <https://about.fb.com/news/2024/04/metasploit-approach-to-labeling-ai-generated-content-and-manipulated-media/>

## Disclosure statement

No potential conflict of interest was reported by the author(s).

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